

ANNUAL REPORT



National Academy of
Agricultural Research Management
Rajendranagar, Hyderabad - 500 030, India

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ANNUAL REPORT

1995 - 96



**National Academy of Agricultural
Research Management**
Rajendranagar, Hyderabad - 500 030, India

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INTRODUCTION

The productivity of National Agricultural Research System is largely dependent on the functional activities and contributions of scientists need to be sensitized to the priorities in agricultural research, which need to be attended to by following mission-mode approach. Personal traits and attributes of scientists play an important role in this activity. It has been amply demonstrated that these traits can be considerably influenced and moulded by appropriate need-oriented training. Since ICAR has large scientific manpower that needs this orientation on continuing basis, it was proposed to establish a Central Staff College for Agriculture in 1976 with the responsibility of training various levels of scientists on the components of research management including project formulation and implementation, accountability in interdisciplinary research project activities, individual and group behaviour, financial and administrative management, and education technology, and for taking up consultancy services in the mandate areas. The Central Staff College was renamed and restructured as the National Academy of Agricultural Research Management (NAARM).

In the short period of its existence, the Academy has trained 6,200 members through 190 training programmes, and 1900 members participated in 60 workshops and conferences. In addition to the national programs, the Academy organized 15 international programmes in which 330 foreign delegates participated.

Considering the need for training the scientists of ICAR and SAUs, several specialized and advanced programmes have been designed for the benefit of middle and senior level research managers as well as for active researchers. In addition to these courses, the Academy also undertakes need-based faculty development programmes for the agricultural universities and other organizations in their respective locations.

The Academy is located at Rajendranagar, a suburb of Hyderabad, near Andhra Pradesh Agricultural University (APAU), about 15 km south of the city.

Objectives of the National Academy of Agricultural Research Management

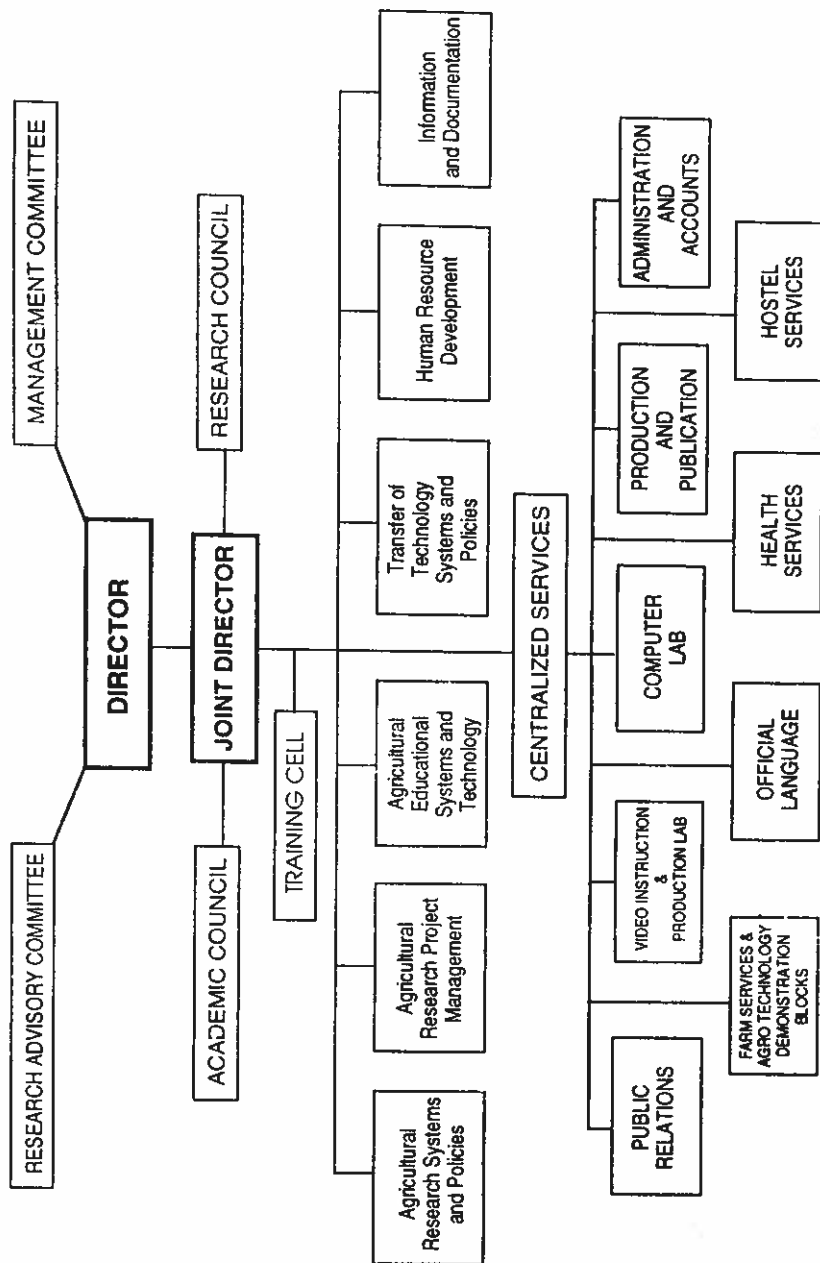
- ✦ To organize foundation courses for the new entrants of scientists to the Agricultural Research Service of ICAR.
- ✦ To develop and organize training programmes in Agricultural Research and Education management for the senior and middle level scientists of NARS.
- ✦ To play a major role in different processes of human resource development of NARS in the context of agricultural research and education.
- ✦ To undertake policy level studies on prioritization of research, resource allocation, and enhancing research productivity in NARS.
- ✦ To evolve suitable management techniques for effective planning, organizing, monitoring, and evaluation of agricultural research projects in NARS.
- ✦ To study various programmes and patterns of technology transfer, and to develop effective systems and strategies for effective communication of agricultural technologies.
- ✦ To undertake developmental programmes for evolving more effective and efficient approaches, methods, and models in higher agricultural education, and to provide support to SAUs and ICAR Institutes in agricultural education.
- ✦ To promote and execute academic programmes of higher learning in specialized areas of agricultural research and education management.
- ✦ To develop expertise in information technology to meet the application software needs of NARS.
- ✦ To undertake systematic review and develop effective systems for efficient administrative and financial management in NARS.

- To function as a repository of ideas in the field of agricultural research and education management, and to act as a clearing house for information dissemination through national and international networks in these areas.
- To offer consultancy services in the field of agricultural research and education management, agricultural information and communication technology.

ORGANIZATIONAL STRUCTURE AND SALIENT ACTIVITIES

The Academy is organized into six subject matter-oriented functional units with a view of impart effective training and to conduct research studies in a meaningful way. These are: Agricultural Research Systems and Policies, Educational Systems and Technology, Transfer of Technology Systems and Policies, Project Management, Human Resources Development and Information and Documentation Units. All the activities of the Academy are supported by centralized services such as Administration and Accounts, Production and Publications, Computer Lab, Video Instruction and Production Lab, Public Relations, Farm Services, Official Language, Health Services and Hostel Services. Presently, the Academy has 168 members including all categories of staff, and senior and principal scientists.

ORGANIZATIONAL STRUCTURE OF NAARM



GLIMPSES

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Achieving success demands total effort. Each step forward will lead us to greater success. The faculty members and staff have all along, endeavoured to develop and maintain the Academy as a centre of excellence. The Glimpses speak for the sustained pace of growth and development of the Academy during 1995-96.

★ A ten-day course on "Educational Video Production" was organized from May 16 to 27, 1995. Twelve participants from various fields of video and publicity, who are engaged in the production of educational video materials, participated in this course.

★ A Faculty Development Programme on "Educational Technology" for the faculty of G.B.Pant University of Agriculture & Technology, Pantnagar was organized from May 18 to 20, 1995 at Pantnagar.

★ The First Phase of the Faculty Development Programme on "Computer Applications" for the faculty members of Andhra Pradesh Agricultural University was organized from May 21 to June 3, 1995.

★ A Summer Institute on "Application of Innovative Instructional Technologies for Optimizing Learning-Teaching in Agricultural Education" was organized from June 5 to 24, 1995. Twenty eight participants in the rank of professors and associate professors from SAUs and ICAR institutes participated in the programme.

★ A one-week course on "Improving the Administrative Efficiency in Agricultural Research Teaching and Extension Organizations of ICAR" was organized from June 14 to 21, 1995. Fifteen senior administrative officers, section officers, and undersecretaries of ICAR headquarters, and assistant administrative officers of various ICAR institutes participated in this course.

★ The second phase of the Faculty Development Programme on "Computer Applications" for the faculty members of Andhra Pradesh Agricultural University was organized from June 19 to 30, 1995.

★ A four-day national workshop on "Poultry, Fisheries and Food Processing" for the Hon'ble Members of Parliament was organized from July 4 to 7, 1995. Fourteen

HOSTEL
SERVICES

HEALTH
SERVICES

OFFICIAL
LANGUAGE

FARM SERVICES &
AGRO TECHNOLOGY
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MPs from Lok Sabha and Rajya Sabha participated. The MPs included Shri Vadde Sobhanadreeswara Rao, Shri Som Pal, Shri Indra Karan Reddy, Shri Dhananjay Kumar, Shri S.P.Gautam, Shri Balayogi, Shri A.C.Dass, Shri Shivcharan Singh, Shri Ram Avadh, Shri Hanumanthappa, Shri U.Venkateswarlu, Shri Reddalah Yadav, Shri S.D.Singh and Shri Devji Kandale.

★ A ten-day course on "Computer Applications in Agricultural Research" was organized from July 11 to 22, 1995. Eighteen scientists from ICAR institutes participated in this course.

★ A ten-day course on "Agricultural Research Project Management" (CARPM) was organized from July 26 to August 5, 1995. Twenty one senior scientists and principal scientists of ICAR institutes participated in this course.

★ CJSC meeting of the ICAR was held at the Academy from August 6 to 8, 1995 under the chairmanship of Dr R.S. Paroda, Director General, ICAR.

★ A three-day national workshop on "Management of Agricultural Research Stations (II. Research Management)" was organized from August 24 to 26, 1995. Dr R.S.Paroda, Director General, ICAR inaugurated the workshop. Fifty Seven delegates comprising Vice Chancellors, Directors of Research, Associate Directors of Research from SAUs, Asst. Directors General of ICAR and Directors of ICAR institutes participated in the workshop.

★ Hindi Fortnight was celebrated from September 1 to 14, 1995. On this occasion, "Krishi Siksha Surabhi (Hindi)" a half-yearly journal on agricultural education and policy was released.

★ An international training programme on "Human Resources Development" was organized for Ms J.O.Fabode, Officer in-charge of Human Resources Development, NARP, Ibadan, Nigeria, from September 4 to 30, 1995.

★ A ten-day training programme on "Training of Trainers" was organized from September 5 to 16, 1995. Twenty three senior level trainers from ICAR institutes, SAUs, Trainers' Training Centres, Extension Education Institutes, and Krishi Vigyan Kendras participated in this programme.

★ A five-day course on "Sericulture - R & D Management" was organized for the Directors of Central Silk Board from September 13 to 16, 1995. The course had a participation of seven Directors from various Institutes.

★ A ten-day course on "Manpower Planning in Agriculture" was organized from September 14 to 23, 1995. Twenty eight participants comprising an Intermix of re-

search and administrative personnel from SAUs, ICAR Institutes, and State departments participated in the course.

★ A five-day training-cum-workshop on "Participatory Methods for Identification and Diagnosis of R & D problems" was organized from September 26 to 30, 1995. Twenty four senior level faculty members and scientists of SAUs and ICAR Institutes including KVKs, and Development Officials of NGOs participated in this programme.

★ An advanced course on "Management of Agricultural Research Stations" was organized from October 26 to November 1, 1995. Twenty six delegates comprising principal scientists, Associate Directors of Research, and Professors/ Station-in-charges of Agricultural Research Stations from various SAUs participated in the programme.

★ An advanced course on "Human Resources Management in Agriculture" was organized from November 8 to 15, 1995. Eight participants comprising principal scientists and senior scientists of ICAR institutes; faculty members from SAUs, and officials from State Departments attended the programme.

★ "Rajbhasha Patraakaarita : Krishi Anusandhan Sansthanoh mei patra patrikaon ko rochak banane ke upay" was organized from November 13 to 17, 1995. Fifteen participants from ICAR institutes and SAUs attended the workshop.

★ A course on "Management for Improving and Assessing Agricultural Research Productivity" was organized from November 14 to 30, 1995. Nine participants in the rank of senior scientists from ICAR Institutes, Joint Directors from Central Silk Board, and professors from SAUs participated in the course.

★ A course on "Sericulture Research Project Management" was organized from November 16 to 25, 1995. Twenty eight Joint Directors and Deputy Directors of Central Silk Board participated in the course.

★ A course on "Scientific and Technical Paper Writing" was organized from November 17 to 23, 1995, to the scientists of ICAR institutes and SAUs. The course was attended by nineteen participants.

★ A course on "Agricultural Scientists Development for Personal and Organizational Effectiveness" was organized from December 5 to 16, 1995. Eleven participants in the rank of professors, associate professors, and senior scientists, from SAUs and ICAR institutes participated in the programme.

★ A workshop on 'Impact of National Agricultural Research Project' was organized from December 14 to 16, 1995. Twenty five participants in the rank of Directors

of Research and Assoc. Directors of Research from SAUs participated in the Workshop.

★ As a part of the one year postgraduate diploma programme in Educational Technology, the second contact session of the Third Outreach Programme in Educational Technology (OPET) was conducted from December 14 to 24, 1995. Twenty one faculty members from different SAUs and ICAR deemed universities participated in the programme.

★ A faculty development programme on " Educational Technology " was organised for the faculty of Sri Venkateswara University, Tirupathi, from January 2 to 13, 1996.

★ A training programme on "Technological forecasting and technical assessment" was organized from January 16 to 24, 1996. A total of twelve faculty members and scientists of ICAR institutes and SAUs participated in the course.

★ A course on " Information Management for Agricultural Scientists" was organized from February 23 to March 1, 1996. Nine scientists from ICAR institutes, SAUs and KVKs participated in the course.

★ A manual on computerisation of accounts titled "Agricultural Research Financial Information System" was published by the Academy. The manual is very handy to all those working on the computerisation of accounts. It was released in the month of February, 1996 at the Directors' Conference held at ICAR, New Delhi.

★ A course on " Video Production and public Relations" was organized to the Mass Communication and Journalism students of Sri Padmavathi Mahila Viswavidyalayam, Tirupathi from March 11 to 16, 1996. Sixteen students participated in this programme.

PROGRAMMES AND ACTIVITIES

3

Academy has been planning, organizing and conducting various training programmes on different aspects of management relating to agricultural research, education, extension education, and training for the junior, middle and senior level scientists of ICAR institutes and State Agricultural Universities. It also conducts training programmes for executives, senior research managers and administrative and accounts officers. Research studies carried out by the faculty members on the functioning of the research and educational institutions in the country provide realistic data for supporting the Academy's programmes and for identifying the management problems including policy issues.

3.1 Training

Training is an important input for Human Resources Development in all categories of human resource-mix engaged in teaching, research and extension functions. Agricultural research and education management is one such important sub-sectors. The Academy imparts training in the areas of research management, educational technology, and training management for various levels of scientists of the ICAR and the SAUs, which form a part of the National Agricultural Research System (NARS). It also conducts training programmes for executives, senior research managers, and administrative and accounts officers. The training programmes have been grouped into two main sub-heads, viz. Foundation course for Agricultural Research Service (FOCARS) for newly recruited Agricultural Research Service (ARS) Scientists, Specialized and advanced training programmes including workshops, seminars, and conferences for senior level scientists and policy makers.

3.1.1 Foundation Course

Foundation course is designed for the scientists recruited to ARS of the ICAR. It is the most important programme of the Academy. The course aims at providing exposure to the trainees on the concepts and principles of project management with special emphasis on project formulation and implementation. It also includes instruction in related areas on human resources development and management, transfer of technology systems and policies, and information management. The specific objec-

tives of the FOCARS are as follows:

Objectives

- + To acquaint the scientist trainees with the general agricultural scenario in the country with specific reference to research, education, extension education and training.
- + To make them aware of the global agricultural scenario in general and international agencies devoted to agricultural development in particular.
- + To orient the scientist trainees in the administrative and management aspects of the Indian Council of Agricultural Research including Agricultural Research Service, Agricultural Scientists Recruitment Board, functioning of various ICAR bodies, personnel policy, finance and monitoring.
- + To expose them to the concept and techniques in agricultural research management in relation to research identification, prioritisation, project formulation, implementation and management.
- + To give an insight into personality development, leadership styles, creativity, problem solving techniques, group processes and conflict management.
- + To provide an understanding and skills in information and communication technology management including computer application, communication aids and techniques, video programme production, skills for oral communication, scientific writing and presentation.
- + To give the trainees an understanding of and an insight into the socio-economic structure of rural society as well as farming community.
- + To expose scientist trainees to the outside world through special scientific and general lectures and seminars, with a view to inculcate social, cultural, moral, and spiritual values, and commitment to work, which help in the creation of a healthy work environment.

The course is structured into three phases, each covering a duration of seven weeks, and extending to a total period of five months. The first phase of the training is at the Academy, and concentrates mainly on the basics of Agricultural Research Management, Organizational Behaviour, Transfer of Technology, Information Management, Administration and Accounts Procedures. The second phase, called Field Experience Training (FET), is held at the regional stations of National Agricultural Research

Project (NARP) of the ICAR Institutes and State Agricultural Universities (SAUs). The FET is designed to impart training in project identification and formulation. This phase helps the trainees to develop an appreciation for rural life and environment. In the third phase of the training at the Academy, the trainees are given an exposure in those areas which would strengthen their skills in scientific writing, and presentation and improvement of communication skills.

The overall programme also includes educational visit to various research institutes in and around Hyderabad to familiarize them with different organizations and the research activities. During the period under report, 80 ARS Scientists, freshly recruited into the service, completed their training, and 45 ARS Scientists were in the process of receiving training. The details are as follows:

S No.	Title of the programme	Period	Co-ordinators	No. of participants
1	FOCARS 52	03.04.95 to 02.09.95	Dr V. Raghavendra Rao Dr K. M. Reddy	19
2	53	18.08.95 to 17.01.96	Dr Jagannadham Challa Dr A. Gopalam	16
3	54	23.01.96 to 20.06.96	Dr P. Manikandan Dr M.M. Anwer	45
4	55	22.03.96 to 20.08.96	Dr A.U. Gadewar Dr P. Kumar	45

3.1.2 Specialized and Advanced Courses

Research management training is a continuous process of learning and of updating skills. The Academy offers various specialized programmes to all categories of scientists. These specialized programmes essentially aim at developing an appreciation among and providing adequate skills to various levels of scientists of the ICAR and SAUs. The trainees of senior courses include Principal Scientists, Professors, Heads of Divisions/Departments, Directors of Research, Project Co-ordinators, Directors of institutes and Deputy Directors General. During the year under report, the faculty had to their credit a record of over 21 programmes, including four faculty development programmes, which are listed in table; and a brief account of each of the programmes is provided below :

Educational Video Production

The course aims at exposing the participants to the concepts of educational video programmes, and the creative aspects of them. It also helps to develop the skills for programme production, and orients the participants to the new and emerging areas of video application.

Computer Applications in Agricultural Research

The programme was organized to expose the participants to the basics of computers, word processing packages such as Wordstar and Microsoft, and other packages like D-Base and Harvard Graphics.

Improving the Administrative Efficiency in Agricultural Research, Teaching and Extension Organizations of ICAR

The course was organized with the purpose of enhancing the skills and efficiency of officers of combined cadre, thereby paving way for constructive administrative support in meeting the organizational goals.

Agricultural Research Project Management

The course exposes the research managers to the applications of management concepts to agricultural research projects for their efficient implementation and also for making them relevant. Various aspects of research project management like project cycle, project formulation, appraisal, budgeting, monitoring and evaluation, management of differences, and brainstorming were covered through information sharing, group discussion, case analysis and exercises.

International Training Programme on Human Resources Development

The programme was organized, for a Nigerian trainee, focusing on aspects like management issues of agricultural research, management information system, Indian experience with the National Agricultural Research Project (NARP), adaptive research, international perspective to management, etc.

Training of Trainers

The programme was organized with a view to helping the teachers and trainees to improve their knowledge and develop certain specialized skills in order to organize and conduct training programmes more effectively. The programme consisted of a blend of lectures, exercises, group discussions, and practice sessions to cover various aspects such as basics of training, role of trainers, training design, delivery, evaluation, etc.

Sericulture R & D Management

The programme was organized to expose the Directors of Central Silk Board to the applications of project management techniques for R & D in sericulture, management information system (MIS), project monitoring and evaluation of sericulture research projects, critical path methods for cost controlling, and computer applications in sericulture research.

Manpower Planning in Agriculture

The course aimed at imparting skills to organize, plan and utilize efficiently the manpower engaged in research, teaching, extension and administration to achieve the objectives of the agricultural research organization. The course covered details on manpower planning, time management, manpower utilization, etc.

Management of Agricultural Research Stations

The programme was organized to expose the participants to the concepts of management of agricultural research stations (MARS) and also to the application of management approaches to these stations. The course covered various aspects like an overview of management of agricultural research stations, establishment of research stations and development, organization models of regional agricultural research stations (RARS), resource management, human transactions in MARS, research project planning and appraisal, research productivity and team work in experimental stations, etc.

Specialized and Advanced Courses Organized

S. No.	Name of the training programme	Name of the co-ordinators	Period	No. of Participants
1.	Faculty Development Programme in Educational Technology at G.B. Pant University, Pantnagar	Dr A. Gopalam	May 8-20, 1995	28
2.	Educational Video Production	Mr K.R.Prabhakar	May 16-27, 1995	13
3.	FDP on Computer Applications at APAU, Hyderabad	Mr U. Muralidhar Dr K.V.S. Rao	May 22- June 3, 1995	20
4.	Improving the Administrative Efficiency in Agricultural Research, Teaching and Extension Organizations of ICAR	Mr M. Suresh Kumar Mr B. Pandu Reddy	June 14-21, 1995	15
5.	Computer Applications in Agricultural Research	Dr K.V.S. Rao Mr U. Muralidhar	July 11-21, 1995	18
6.	Agricultural Research Project Management	Dr K. Venkateswarlu Dr S.N. Saha	July 26 - Aug 5, 1995	21
7.	International Training Programme on Human Resource Development	Dr J. Challa	Sept. 3 - 23, 1995	1
8.	Training of Trainers	Dr T. Balaguru Dr P. Manikandan	Sept. 5 - 16, 1995	23
9.	Sericulture R & D Management	Dr K. Venkateswarlu	Sept. 13 - 16, 1995	7
10.	Manpower Planning in Agriculture	Dr VE. Sabarathnam	Sept. 14 - 23, 1995	28
11.	Management of Agricultural Research Stations	Dr K. Venkateswarlu Dr J.C. Kalla	Oct. 26-Nov. 1 1995	23

No. of Participants	
5	28
95	13
3,	20
5	
995	15
95	18
5,	21
5	
1995	1
1995	23
1995	7
1, 1995	28
1 1995	23

S. No.	Name of the Training Programme	Name of the Co-ordinators	Period	No. of Participants
12.	Human Resource Management in Agriculture	Dr M.M. Anwer	Nov. 8-15, 1995	8
13.	Management of Improving and Assessing Agricultural Research Productivity	Dr P. Kumar Dr V.R. Rao Dr VE Sabarathnam	Nov. 14-20, 1995	9
14.	Sericulture Research Project Management	Dr K. Venkateswarlu Dr J.C. Kalle	Nov. 16-25, 1995	28
15.	Scientific and Technical Paper Writing	Dr S.N. Saha Dr P. Manikandan Ms.M.Suguna Sri	Nov. 17-23, 1995	19
16.	Agricultural Scientist Development for Personal and Organizational Effectiveness	Dr S. Shanmugam Dr J. Challa	Dec. 5-16, 1995	11
17.	FDP in Educational Technology at Sri Padmavathi Mahila Viswa Vidyalaya, Tirupathi	Dr A. Gopalam	Jan. 2-13, 1996	28
18.	Technological Forecasting and Technological Assessment	Dr. D. Rama Rao	Jan. 16-24, 1996	12
19.	Information Management for Agricultural Scientists	Ms.M. Suguna Sri Dr K.V.S. Rao	Feb. 23 - Mar. 1, 1996	9
20.	Video Production and Public Relations	Mr. K.R. Prabhakar Mr A. Bhattacharjee Mr R.V.V.S. Prakasa Rao	Mar. 11-16, 1996	16
21.	FDP in Educational Technology (AHRD) at Madras	Dr A. Gopalam	Mar.11-20, 1996	35

Human Resource Management in Agriculture

The course was organized to expose the scientists to the concept of human behaviour in organizations and to examine the issues of management of people in agricultural research, which, in turn, would help to improve the research productivity in the organizations. It covered various areas which included organizational behaviour, values and attitudes, personality and stress, perception and motivation, fundamentals of group behaviour, group dynamics, leadership, conflict management, organizational structure, performance evaluation and rewards, etc.

Management of Improving and Assessing Agricultural Research Productivity

The course was organized to improve the research productivity of the agricultural scientists working in ICAR research institutes and SAUs, by enabling them to understand and practice the various skills needed for improving the research performance and productivity. It covered a variety of topics which have a direct bearing on the research productivity. Case studies, experiences of various countries, and practical management theories were discussed in the context of Indian agricultural research.

Sericulture Research Project Management

The course aimed at exposing the participants to the concept of research project management in the context of sericulture research, and also to make them understand the management of material, manpower, money, and time in managing specific research projects. It covered project planning, selection, creativity and problem solving, systems approach in project planning, project budgeting, and project monitoring and evaluation.

Scientific and Technical Paper Writing

The course aimed at exposing the participants to the need and importance of effective scientific writing, and also to develop skills in the participants for effective scientific writing. It provided a good forum to the scientists for discussing various aspects of scientific and technical writing.

Agricultural Scientist Development for Personal and Organizational Effectiveness

The course was organized to familiarize the scientists with the concepts and skills for enhancing personal and organizational effectiveness. It covered HRD strate-

gies for agricultural research, personal effectiveness as a scientist, motivation of scientists at work, interpersonal relationships and transactional analysis, team work and leadership, managerial grid, management of differences, organizational communication, administrator-scientist interface, and organizational development.

Technological Forecasting and Assessment

The programme has been designed to train the scientists from research and development institutions in the principles of technological forecasting. The programme discussed the case studies in agriculture, veterinary and engineering disciplines developed by various experts in the country.

Information Management for Agricultural Scientists

The programme was organized to familiarize the participants with new concepts, technologies, products and services for managing information effectively and to develop skills in handling information collection, organization, retrieval, repackaging and dissemination.

Video Production and Public Relations

The course was organized to acquaint the participants with the latest developments and the modern techniques in video production, and facilitate them to acquire specialized knowledge in the areas of printing, editing and public relations activities.

3.2 Research

One of the important mandates of the Academy is to conduct research in the area of agricultural research management. It carries-out research in various areas such as agricultural policy, research project management, transfer of technology, human resources development, agricultural education, and information technology. A brief description of the research projects is presented below :

A profile of scientific human resources in ICAR system in India (Dr S. Shanmugam)

The project was undertaken to study the socio-economic attributes of cadre strength of scientific personnel in ICAR Institutes, and to plan the optimum human resources profile in ICAR System for 2001 A.D.

The information about the ICAR scientists upto 1993 were collected and analysed. Updation of data is under progress.

Information generation and information management in KVK - A comparative analysis

(Dr A.U. Gadewar and Dr K. Vidyasagar Rao)

The research project was undertaken to study the information management in Krishi Vigyan Kendras. A questionnaire was designed and developed to collect preliminary information on KVKs.

Scope for privatising farm extension - An ex-ante analysis

(Dr Rasheed Sulaiman and Dr A.U. Gadewar)

The project was initiated to investigate the role of public, co-operative and private organizations, and other information services engaged in extension / transfer of technology; to ascertain the effectiveness of alternative extension arrangements in terms of subject-matter content, linkage, planning, clients and costs; to understand the capacity and willingness of farmers to pay for various extension services and the probable mechanisms of payment; to provide policy level input and guidance, and to suggest structural changes in the organizational set up for appropriate combinations of different agencies / institutions engaged in agricultural extension / transfer of technology.

Developing suitable Management Information System (MIS) for NARS

(Dr T. Balaguru, Dr P. Manikandan, Dr Jagdeesh C. Kalla and Dr K. Vidyasagar Rao)

The project was taken up to develop a methodology for the development and use of management information system to improve the effectiveness of research through better decision-making regarding the utilization of scarce research resources.

The basic MIS methodology was published in the form of a book for use by the ICAR scientists and agricultural university teachers in developing, maintaining and making use of a computer-based MIS in their respective institutions.

Evolving a suitable methodological framework for priority setting in agriculture

(Dr T. Balaguru)

The project was initiated with the objective of developing suitable methodology for setting research priorities in agriculture. For this purpose, some of the formal methods for prioritizing research at the national and institute levels were screened. Basic data collection from the scientists of a few selected institutes in the NARS was initiated.

Development and analysis of database on crops, forestry, livestock and fishery resources in NARS

(Dr K. Vidyasagar Rao and Dr P. Kumar)

The data collected earlier on livestock population and productivity were analysed, and interpreted during the year under report. The trends in livestock population and productivity were studied for the past fifteen years utilizing the secondary data. The period of study was divided into three parts of five years each (1980-84, 1985-89 and 1990-94).

The mean, standard deviation and percentage increase over the previous period were calculated to study the trends. The research findings are under print.

Technological forecasting of groundnut scenario in India

(Dr D. Rama Rao and Dr Kiresur)

The objective of the project is to forecast groundnut scenario in the country by 2000 AD using technological forecasting methods.

The secondary data were collected mainly from various publications of Directorate of Oilseeds Research, Hyderabad; Oilseeds Scenario by CERPA, Delhi, and from other research journals and books. The historic data were used to prepare a simple economic forecast. Experts opinions were obtained through Delphi on various issues of groundnut. The final report was prepared using economic forecast coupled with the Delphi analysis.

Professional (graduate) women in agriculture (PWA)

(Dr D. Rama Rao and Ms M. Suguna Sri)

The Project was undertaken to study the status and growth of agricultural women graduates; career opportunities for agricultural women graduates; professional performance of women graduates; and identification of issues pertaining to women professionals, and the influencing factors.

During the period under report, fifteen experts were invited to participate in the project as PAC members. The first PAC meeting was held to review the project objectives, methodology, technical programme and future actions. It was decided to execute the project with active support from resource persons from 28 SAUs. The VCs of the SAUs have been asked to nominate a lady faculty as a resource person for the project.

Three different questionnaires were designed. The first one is for the organizations to seek information on women graduates employed. The second questionnaire is for the academic institutions to collect data on students. The third one is for seeking the opinions of individual graduate women through a survey.

Management and impact of agricultural research stations /institutes in India

(Dr. K. Venkateswarlu)

The project aims at studying the processes and details of technology developed in agricultural research stations and also the production and productivity of various research / experimental stations. The constraints faced in the management of research stations would also be found out. Case studies have been initiated and data are being collected for this purpose.

Management and impact of All India Co-ordinated Research Projects

(Dr. K. Venkateswarlu and Dr. S. Shanmugam)

The major objectives of the project are to identify the lacunae in the structure and organization of the All India Co-ordinated Research Projects (AICRPs) and also to

tions of Directo-
Delhi, and from
prepare a simple
on various issues
coupled with the

study the impact of various AICRPs. A questionnaire has already been designed and mailed to the project co-ordinators to collect the data on these aspects. The data collection is in progress.

Impact of National Agricultural Research Project

(Dr. K. Venkateswarlu and Dr. K. Vidyasagar Rao)

The project is taken up to develop a statistical model for measuring the impact of National Agricultural Research Project (NARP) in 120 agro-climatic zones. The data collection is in progress.

Criteria for selection and formulation of R&D projects in agriculture and in allied areas

(Dr. K. Venkateswarlu)

gricultural women
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to women pro-

to participate in
view the project
was decided to
28 SAUs. The
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The project aims to study the criteria for selection of R&D projects in agriculture and in allied areas, and also to study the formulation process of various R & D projects. Data are being collected from various organizations like ICAR, SAU, CSIR, CSB, CTB, ICMR, ISRO, IIM, BARC, etc., for this purpose.

Creativity and problem-solving nature of agricultural scientists

(Dr. K. Venkateswarlu)

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The main objective of this study is to understand the creativity and problem-solving nature of scientists in developing new theories / inventions / discoveries in the field of agricultural sciences. The data have been collected from the scientists and senior scientists through a standard questionnaire. The data are yet to be collected from top level scientists and research managers. The project is under progress.

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3.3 Education

Summer Institute on Application of innovative instructional technologies for optimizing learning-teaching in agricultural education

The Summer Institute on "Application of innovative instructional technologies for optimizing learning-teaching in agricultural education" was organized from June 5 to 24, 1995. The programme aimed at exposing the faculty members of both the agricultural universities and the deemed-to-be universities of ICAR to the various instructional systems, methods and strategies which, in turn, would help in enhancing the teaching-learning effectiveness in agricultural education.

The programme centred around the basic theme that instruction is both science and art. The scientific side of instruction takes into account the methodology and principles, whereas the art component emphasizes on the management of motivation, learning behaviour, and personality of teachers. The programme employed a variety of techniques to achieve its objectives. In addition to exposing the participants to the latest theories related to teaching-learning, the programme had incorporated in its curriculum computer and video-based individualized instructional techniques, group learning processes for optimizing instructional effectiveness, case studies, role plays, etc., all of which helped in developing adequate skills in the participants to improve the learning-teaching effectiveness. The concept of vocationalization in agricultural education was also discussed during the course of the programme, keeping in view the larger role of instruction in dealing with agricultural education.

Twenty eight participants in the rank of professors and associate professors from SAUs and ICAR Institutes attended the Summer Institute, for which Dr A. Gopalani was the Director.

Educational video production in agriculture and allied topics

Video Instruction and Production Lab (VIPL) has produced 35 video programmes on various topics, which were prepared and presented by various scientists who underwent training at NAARM during the period under report. Some selected programmes include

- Crab farming
- Perspectives of sustainable agriculture
- Organic farming
- Production and use of trichogrammatid egg parasites

Salt affected soils
 Pesticides - Do's and don'ts
 Health and quiz vitamin
 What is a computer ?
 Isolation distance in hybrid rice
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Outreach Programme in Educational Technology

The Academy offers one-year postgraduate diploma programme in Educational Technology (OPET). The OPET programme, even though offered based on the philosophy of distance education, is different from the other conventional distance education programmes where only print medium is used as carriers of instruction. This programme facilitates incorporation of essentials of educational technology in the agricultural university education. The principle objective of the OPET is to offer a orientation programme to all faculty members of State Agricultural Universities and ICAR deemed Universities covering all aspects of higher agricultural education, more specifically, in educational methodology and instructional technology to produce appropriately trained persons, who could serve the needs of these institutions by way of organizing in-service training programmes for teachers, undertaking education research, contributing to the upgradation of educational programmes, learning-teaching activities, etc., and by encouraging the agricultural university teachers to conduct research in learning-teaching, testing, evaluation, and other allied fields of agricultural education.

After the successful completion of two such programmes, the third OPET programme was initiated with 22 participants. For each programme, there will be three contact sessions - one week of orientation in the beginning, two weeks in the middle and another two weeks at the end. These contact sessions are compulsory, and are meant to train in instructional skills, preparation of resource materials like audio tutorials, computer assisted programmed lessons, self-instructional modules, evaluation materials, etc.

During the year under report, the second contact session of the third OPET was conducted from December 14 to 24, 1995. Twenty two academicians of various cadres from SAUs attended this programme, co-ordinated by Dr A. Gopalam and Dr K.M. Reddy.

3.4 Workshops

To facilitate the exchange of information and interaction among the scientists of the National Agricultural Research System, the Academy conducts a number of workshops. These workshops address the current and emerging areas of interest in the field of agricultural research management. During the year under report four workshops were conducted, the details of which are as follows

Workshops organized

S. No.	Name of the workshop	Period	Co-ordinators	No. of participants
1.	Poultry, Fisheries and Food Processing for the Hon'ble Members of parliament	July 4 to 7, 1995	Dr J.C. Kalla Dr P. Kumar Dr S. Shanmugam	14
2.	Management of Agricultural Research Stations	Aug. 24 - 26, 1995	Dr K. Venkateswarlu Dr J.C. Kalla	62
3.	Participatory Methods for Identification and Diagnosis of R & D Problems	Sept. 26 - 30, 1995	Dr A.U. Gadewar Dr J.C. Kalla	24
4.	Impact of National Agric. Research Project	Dec. 14 - 16, 1995	Dr S.N. Saha Dr K. Venkateswarlu	31

National Workshop on Poultry, Fisheries and Food Processing for the Hon'ble Members of Parliament

The workshop was a very senior-level programme conceptualized by the Ministry of Parliamentary Affairs, involving seven ministries. Fourteen MPs from Lok Sabha and Rajya Sabha participated. The MPs included Shri Vadde Sobhanadeeswara Rao, Shri Som Pal, Shri Indra Karan Reddy, Shri Dhananjay Kumar, Shri S.P. Gautam, Shri Balayogi, Shri A.C. Dass, Shri Shivcharan Singh, Shri Ram Avadh, Shri Hanumanthappa, Shri U. Venkateswarlu, Shri Reddaiah Yadav, Shri S.D. Singh and Shri Devji Kandale. Mr. Shivcharan Singh, MP inaugurated the exhibition which was organized to display the latest breeds in poultry and fisheries.

The workshop provided a common platform for deliberating on the thrust areas which are crucial for the overall integrated agricultural development of the country.

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Mr. P. Hanumanthappa, MP, lighting the lamp to mark the inauguration. Dr M.V. Rao, VC, APAU; Dr Jayadeesh C. Kalla, Director, NAARM, and Dr Krishna Rao, former VC, APAU are also seen.



A view of the Members of Parliament in the workshop on poultry, fisheries and food processing.

Several eminent resource persons from the fields of poultry, fisheries, food processing and extension took part in the workshop.

The workshop was co-ordinated by Dr Jagdeesh C. Kalla, Dr P. Kumar and Dr S. Sharmugam. Dr Jagannadham Challa acted as the organizing secretary.

Workshop on Management of Agricultural Research Stations

The workshop was organized to facilitate exchange of practical experiences on management of agricultural research stations through case studies developed for ICAR Institutes, Regional Agricultural Research Stations and the State Agricultural Universities. This workshop was a continuation of the earlier workshops conducted at the Academy on the management of agricultural research stations. In this workshop, various aspects related to research management in the agricultural research stations were deliberated and discussed. A total of 62 delegates, comprising Vice Chancellors, Directors of Research, Associate Directors of Research from SAUs, and Assistant Directors General and Directors from the ICAR system, participated in the workshop. The workshop was inaugurated by Dr R.S. Paroda, Director General, ICAR, New Delhi.

Training-cum-workshop on Participatory Methods for Identification and Diagnosis of Research and Development Problems".

The workshop aimed at providing an understanding of various participatory methods, and at designing and developing tools and techniques of Participatory Technology Development (PTD) and Participatory Rural Appraisal (PRA). The main objectives of the workshop were

- To acquaint the participants with various participatory methods of identifying and diagnosing various agricultural and rural problems, and to help in designing and developing tools and techniques of PTD and PRA

- To assess the already employed methods and to help in improving one's own work by supplementing with PRA/PTD methods

- To experiment and evaluate alternative methodologies in real appraisal setting

Group discussions on methodological issues and application of PRA in KVKs, watershed management, training programmes and transfer of technology projects were

conducted during the workshop which was attended by senior level faculty members and scientists of SAUs and ICAR Institutes including KVKs, and developmental officials of NGOs.

Some of the recommendations brought out during the workshop include

- PRA techniques with appropriate survey tools can be used for identification of training needs in Krish Vigyan Kendras.
- PRA techniques could be used for watershed programme planning and programme implementation.
- PRA methodology should be incorporated in the curriculum of graduate and postgraduate studies in the agricultural universities.
- PRA is being used in FSR/E. It needs to be institutionalized.

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Dr R.S. Paroda, DG, ICAR, addressing the delegates of the workshop. To his right are Dr K. Venkateswarlu, Co-ordinator of the workshop, Dr Jagdeesh C. Kalla, Director and Dr S. Shanmugam, Sr. Scientist, NAARM.



Dr R.S. Paroda, DG, ICAR in discussion with the delegates of the workshop

3.5 Consultancy

3.5.1 Faculty Development Programmes

Considering the expertise developed at this Academy in the area of educational technology and management, this Academy has undertaken the major HRD activity through faculty development programme in the state agricultural universities and in the four deemed-to-be universities of ICAR. The new package of faculty development programme has been developed for SAUs as per their needs. The following faculty development programmes (FDP) were organized by the Academy during the period under report

Faculty Development Programmes Organized

S. No.	Programme	Period	University
1	FDP in Educational Technology	May 8 - 28, 1995	G.B. Pant University Pantnagar
2	FDP on Computer Applications	May 21 to June 3, 1995	Andhra Pradesh Agric. University, Hyderabad
3	FDP in Educational Technology	Jan. 2 - 13, 1996	Sri Padmavathi Mahila Vishwa Vidyalaya, Tirupati
4	FDP in Educational Technology	Mar. 11 - 20, 1996	TNVASU, Madras

Faculty Development Programmes in Educational Technology

The off-campus programmes in Educational Technology were organized at the request of the respective vice chancellors of the concerned universities. The areas emphasized during these programmes include the concepts of management of agricultural education, behavioural modification in learning and teaching, testing, evaluation, and curriculum development.

Faculty Development Programme in Computer Applications

To sensitize the faculty members of Andhra Pradesh Agricultural University on the computer applications in agricultural education, the programme was organized at the request of its Vice Chancellor in three phases.

During the first phase (from May 21 - June 3, 1995) twenty faculty members were exposed to the basics of computers, word processing packages and other data-base packages.

The twelve-day long second phase of the programme which commenced on June 19, 1995, covered electronic spread sheet and programming in Basic. A blend of lectures, demonstration classes, and practical hands-on experience were adopted to impart training during these two phases.

The third phase was designed in such a way that the learning group had access to the computer resources personnel of the Academy for feedback, and for clarification of doubts, while the learners conducted the course in their respective institutions.

3.5.2 Consultancy to NRCS, Indore

Consultancy was provided to National Research Centre for Soyabean, Indore, for production of video on NRCS. Script has been prepared. Production will be taken up during the next year.

3.5.3 Consultancy to other training institutes

The Academy provided expertise in the form of guest lectures to other local training and educational institutions such as the Extension Educational Institute, the Home Science College of the Andhra Pradesh Agricultural University, SVP National Police Academy, Institute of Co-operative Management, Regional Station for Forage Production and Demonstration, International Crops Research Institute for the Semi-Arid Tropics, WALAMTARI, NISIET, and NIRD. Most of the faculty members of the Academy were invited to deliver guest lectures in the above mentioned institutions. Different areas of management and educational technology were covered in these invited lectures.

3.6 Miscellanea

3.6.1 Progress of Hindi Implementation

Hindi Fortnight Celebrations

Hindi fortnight was celebrated at the Academy from September 1 to 14, 1995. Several competitions like noting, drafting, essay writing, elocution were conducted in Hindi to inculcate Hindi usage among the staff and officers of the Academy.

Dr. Mohan Singh, Vice Principal of the Arts College, Osmania University, Hyderabad, who was the chief guest on the concluding day, honoured the winners with certificates and prizes. *Krishi Siksha Surabhi* (Hindi), a half-yearly journal on "Agricultural education and policy" was released.

The Hindi cell of the Academy extended co-operation and help in organizing the Hindi week celebrations in Directorate of Rice Research and National Research Centre for Sorghum, Hyderabad.

Hindi Software Installed

"Shabd Ratna," a Hindi software package, was installed in all the computers at the Academy to facilitate the staff and officers to work in Hindi.

Workshops Organized

A workshop on "*Rajbhasha Patrakarita: Bharatiya Krishi Anusandhan Parishad evam adhinasth sansthanom ke patrikaon ko ruchikar banane ke sandarbh me*" was organized from November 13 to 17, 1995 for the Hindi officers, translators and Hindi Assistants of ICAR Institutes and other Central Govt. organizations and undertakings.

Another workshop was organized from November 13 to 17, 1995 by Central Hindi Training Sub-Institute, Hyderabad, for the employees of the Academy to impart the skills required for discharging their official duties like correspondence, noting, drafting, etc. in Hindi. Fifteen employees were trained through this workshop. Mr Devidas, Asst. Director, CHTSI, Hyderabad acted as a facilitator of the programme.

3.6.2 Other Important Activities

Sadhbhavana Diwas

The Academy observed Sadhbhavana Diwas on August 20, 1995. The staff members assembled and took a pledge to preserve the harmony and unity of the nation.

Anti-terrorism Day

The Academy observed Anti-terrorism Day on May 21, 1995.

Health Centre Foundation

Dr R.S. Paroda, Director General, ICAR, New Delhi, laid the foundation stone of the NAARM Health Centre building in NAARM Campus on August 8, 1995.

Blood Donation Camp

Blood donation camp was organized by NAARM Health Centre on March 2, 1996, with the help of Indian Red Cross Society, Hyderabad branch and Blood Bank, Govt. Maternity Hospital, Nayapul. Fifteen persons donated blood on that day.

Pulse Polio Immunization Camp

Pulse Polio Immunization Camp was organized at the Health Centre, as a part of mass immunization of all children upto three years age all over India, on December 9, 1995, and on January 20, 1996. The aim of the pulse polio immunization was to eradicate polio from India. Eighty three children were immunized on both the days.

3.6.3 Guest Lectures

The following guest lectures were organized at the Academy for the benefit of trainee scientists, faculty members and staff members of the Academy:

- ** "Scientists vs Bureaucrats" by Prof. E.G. Parameswaran, Osmania University, Hyderabad, on June 21, 1995.
- ** "Stress Management" by Dr Ajit Vigg, Physician and Chest specialist, on June 28, 1995.

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View of the Central Joint Staff Council meeting chaired by Dr R.S. Paroda, DG, ICAR



AIIS probationers in action during the Participatory Rural Appraisal, which forms part of their Field Experience Training

- ** "Missiles" by Lt. Gen. Sundaram, Director, Defence Research and Development Laboratory, on July 12, 1995.
- ** "Use of plain English in agricultural research" by Prof. G.V. Raj, Former Professor, Osmania University, Hyderabad, on August 24, 1995.

3.6.4 Felicitation

The Noble Laureate and the father of green revolution, Dr Norman E. Borlaug, was felicitated at the Academy on Feb. 6, 1996. He called upon the scientists to serve the human needs by ensuring that the strides made in research reach the people. Dr Borlaug said that no nation should become complacent after achieving self sufficiency in foodgrain production. One has to keep pace with the changing times, and the benefits of agricultural revolution should reach the real beneficiaries. Prof. Timothy Reeves, Director General, International Wheat and Maize Development Centre (CIMMYT), who was also present on the occasion, said that more concentration should be laid on productivity. India was doing well in the international market in terms of export of agricultural produce, but it should sustain the competitiveness, he added.

The function was organized by the scientists of the ICAR institutes in Hyderabad, to felicitate Dr. Norman E. Borlaug.

4.1 Facilities

The Academy has good infrastructural facilities for carrying out its various activities. These facilities include.

4.1.1 Video Instruction and Production Laboratory

This Laboratory, comprising two television studios and a Master Control Room, facilitates intensive training of scientists in the preparation of effective resource materials for telecasting. It helps in conducting role play exercises for personality development. Video educational programmes are also produced. It is also used for developing proper communication skills.

4.1.2 Audio-Visual and Photography Laboratory

This laboratory, equipped with all modern photographic and audio visual facilities, provides an excellent opportunity to gain experience in the preparation and use of audio-visual aids for effective presentation. The facilities include overhead projectors, epidiascope, coloured transparency - making units, slide projectors, tape recorders and public address systems, built-in sound slide projector with slide synchronising facility, audio cassettes duplicator, slide dissolve unit system, and laminator machine.

4.1.3 Speech and Mass Communication Laboratories

These Laboratories facilitate the scientists to record their programmes such as interviews, discussions, and talks, specially for the rural listeners.

4.1.4 Organizational Behaviour Laboratories

Individual and group exercises in team building, leadership, and group dynamics, conducted in these laboratories, provide the trainees an opportunity to understand the various aspects of organizational behaviour including human thought, action and behaviour.

4.1.5 Self-aided Instructional Laboratory (SAIL)

This laboratory provides opportunity for the participants of various programmes to undergo structured and programmed instruction under self-learning situations. Automated electronic equipment and computerized instruction facilitate the process. A set of carrels, with Telex Caramate for imparting a tape-slide programme, a VCP with monitor for a video tutorial, and a computer for programmed instruction, are used.

4.1.6 Library

The Academy has a specialized library with its collection focusing on agricultural research management, science policy, system studies, project management, futurology, forecasting, educational technology, technology transfer system/policies, women studies, information management, research communication, computer/information technology, human resource development, administration, etc. The library has a collection of more than 16,000 books, and over 360 books were added during the year. The library subscribes to 120 foreign journals, 100 Indian journals, and gets 60 journals and newsletters on gratis/exchange. Besides this, the library has a good collection of reports and other publications of various international organizations and has also foreign tour reports submitted by the ICAR scientists.

It has a CAT database using CDS/ISIS package ver 3.07 for its book collection and is producing computer printed card catalogue from this database. The CAT database will very soon be made available for patrons with a search interface like Heurisko. The library has also developed an FTR database for accessing the collection of foreign tour reports of ICAR scientists. The library has a CD-ROM work station with a Hitachi CD ROM drive and a DCM-TURBO 2000 PC-AT. The CD-ROM databases available with the unit are AGRIS (1989-1995); AGRICOLA (1980-1994) ; AGRI SEARCH (1990-1994) ; TROPAG AND RURAL (1975-1992).

The General Information Service Terminal of National Informatics Centre (GISTNIC), installed in the library, makes available the basic information about India, and also various important national and international information, viz. agricultural/census/economic statistics, railway/air time table, recreational and tourist information. GISTNIC terminal at the Academy is extensively used for supplying statistical information to the scientist trainees, faculty members and staff of the Academy.

The reprographic section of the Unit provides extensive support to various training programmes and workshops conducted by the Academy. It is equipped with Finnese 4550, UBIX 1802, and NASHUA 7140 xerox machines and RICOH 6950 plain copier.

A secondary publication titled "Agricultural Research Management Abstracts" (ARMA) is brought out by the Information and Documentation Unit.

4.1.7 Computer Laboratory

A computer laboratory has been established with a mini frame computer having 4MB RAM and 300 MB hard drive alongwith 20 personal computers. The system is provided with three dot-matrix printers, two near-letter quality printers of FX Series, and one ink-jet printer. Four PC-AT-486 with 8MB RAM and 200/500 MB hard drive alongwith SVEA colour monitors have been acquired to strengthen the areas like Artificial Intelligence/Expert systems, multimedia, and geographic information system. Two multi-media laptop computers for classroom projection were acquired alongwith fifteen digital 486 computers. Among these two ATs were provided with multi-media kit. Printer sharer, CCD plate for computers, and in video projection have also been provided for effective use of computer practicals, all of which help in better demonstration and in better viewing.

These systems are mainly used to provide intensive training on computers to various levels of scientists from the ICAR and SAU system, who participate in different programmes conducted by the Academy. Softwares like PC Box, Bilingual wordprocessing DOS, WINDOWS, UNIX, CLIPPER, WORD PERFECT, QPRO, LOTUS 1-2-3, REFLEX, dBASE, HARVARD GRAPHICS, PILOT STORY BOARD, and CDS/ISIS, have been loaded in all the personal computers for effective use in training as well as for day-to-day activities. Further, they are put into extensive use for the preparation of databases and development of library information system. The computers are also being used for the administrative work, particularly in preparation of pay bills, accounting and budgeting, recruitment processing, and for the secretarial work. Necessary training is also imparted to the staff on the use of application packages like Wordperfect, dBASE, and Lotus, with the aim to increase the efficiency of their working conditions. There is a proposal to upgrade the Computer Laboratory to take care of the areas like Artificial Intelligence/Expert Systems, Multimedia, and Geographic Information System.

4.2 Services

4.2.1 Administration, Finance and Accounts

This wing of the Academy is organized into five sections : Administration I deals with establishment matters; Administration II with cash and bills; Administration III with stores and purchases; Administration IV with estate management and training; and all matters pertaining to expenditure and bank reconciliation statements, monthly and annual accounts, budget and revised estimates, and internal audit are dealt with by the audit and accounts section.

This wing provides administrative, financial, and accounts support to all the units of the Academy. Further, the Chief Administrative Officer and the Finance and Accounts Officer function as regular faculty members in delivering lectures on various rules and regulations of ICAR to ARS scientists and other trainees.

4.2.2 Farm Services

The Academy has developed the Agro-Technology Demonstration Blocks (ATDB) in an area of 30 hectares, which act as the show-window of the National Agricultural Research System. The concept of ATDB is a unique innovation where the technologies developed and released by different ICAR institutes and SAUs are demonstrated at one place for the benefit of national and international visitors. During the year under report, about 2000 visitors and 300 scientist trainees had an opportunity to view these demonstrations. Lawn mover was purchased during the year under report. Irrigation pipe line was also laid for better irrigation facilities.

4.2.3 Production and Publication Services

The Production and Publication Cell undertakes publication of various materials developed by faculty for training purposes. The cell is engaged in production of books, Annual Report, Folders, hand-outs, leaflets, information bulletins of the Academy, and other items of work like registers, stationery forms, letterheads, cover pages, invitation cards, etc. RICOH 6950 automatic plain paper copier was also acquired to enhance the production activities at a faster pace. The cell is also equipped with Desktop Publishing (DTP) System with several packages including MS Word, MacWrite, MacPaint, MacDraw, Page-Maker and Hindi Font. This cell is equipped with an offset printing press (Swift 300 Delux) a processing camera, and plate making, screen printing, and binding systems. The important publications brought out are NAARM News, Annual Report, NAARM Deepika, Recommendations of various Workshops, and ARMA (Agricultural Research Management Abstracts).

4.2.4 Hostel Services

The Academy has good board and lodging facilities for its visitors and trainees. The Academy's programmes are fully residential, and the training participants are accommodated in the Halls of Residence and in the Scientists Home. These residential facilities have excellent catering services, which provide Indian, Chinese, Mughlai, and Continental cuisine. The Halls of Residence contains 154 fully-furnished single occupancy rooms with attached bath. The Scientists Home has 32 suites. Each suite is furnished with two beds, sitting and study areas, and an attached bath. Half of these suites are air-conditioned. There is a public call office in the Halls of Residence with STD/ISD facility. Colour Television was installed in the dining hall for recreation of the trainee scientists. Believing that "health is wealth", gym equipment was also purchased for the physical exercise purposes for both the trainees as well as for the staff of the Academy.

4.2.5 Banking Service

The Central Bank of India, Golconda Fort Branch, Hyderabad, has been operating an Extension Counter (regular) in the faculty- cum-laboratory building to cater to the banking requirements of faculty members, staff and trainees. The counter also offers safe-deposit locker facility to the faculty members and to the staff members.

4.2.6 Health Services

Medical services are extended by a full-time medical doctor and a compounder, during scheduled hours in the Health Centre of the Academy, to the faculty members, officers, staff, and their families, and to the trainees. Emergency care is provided round-the-clock. ECG analyzer, blood sugar estimator, oxygen instrument, and suction apparatus are available at the centre for extending better medical care and treatment to the patients.

4.2.7 Public Relations Cell

The Public Relations Cell helps to establish effective linkages with the media by focusing attention on the activities and achievements of the Academy. The cell, besides building the image of the Academy, also provides support in the areas of publicity, visitors co-ordination, graphics art, and in arranging various guest lectures. The cell also assists in organizing cultural programmes, field trips, and sightseeing programmes.

4.3 Other facilities

4.3.1 Residential Facilities

The Academy has a total of 95 residential quarters of different categories for the purpose of accommodating scientists, technicians, and staff of NAARM and other ICAR institutions in Hyderabad.

4.3.2 Recreation and Sports Facilities

The recreation facilities provided include several televisions connected to a dish antenna, common rooms with indoor games, and courts for tennis, volley ball and badminton.

TECHNICAL AND OTHER INFORMATION

5.1 Finance and Accounts

NON-PLAN 1995-96

S. No	Head of account	Budget estimates 1995 - 96 (Rs.)	Revised estimates 1995-96 (Rs.)	Final grant (Rs.)	Expenditure during 1995-96 (Rs.)
1.	i) Establishment charges				
	a. Salary of officers				20,17,685
	b. Salary of establishment				22,09,206
	c. Dearness allowance				45,34,200
	d. House rent allowance	1,12,00,000	1,08,50,000	1,08,03,000	5,28,879
	e. City comp. allowance				1,20,583
	f. Other allowance/Hon.				4,58,536
	g. Interim relief				5,41,451
	h. Bonus				3,63,348
	i. Leave salary & Pension contribution				—
	Total	1,12,00,000	1,08,50,000	1,08,03,000	1,07,73,888
	ii) Overtime allowance	21,000	21,000	21,000	20,990
	iii) Travelling allowance	1,60,000	1,60,000	3,29,000	3,17,312
2.	Other charges (excluding major construction works)	29,19,000	49,69,000	49,69,000	49,61,165
	Grand Total	1,43,00,000	1,60,00,000	1,61,22,000	1,60,73,355

	Head of account	Budget estimates 1995 - 96 (Rs.)	Revised estimates 1995-96 (Rs.)	Final grant (Rs.)	Expenditure during 1995-96 (Rs.)
1	i) Establishment charges				
	a. Salary of officers				11,60,880
	b. Salary of establishment				9,578
	c. Dearness allowance				14,74,417
	d. House rent allowance	1,00,00,000	32,50,000	28,00,000	150
	e. City comp. allowance				51,731
	f. Other allowance/Hon.				5,296
	g. Interim relief				75,036
	h. Bonus				-
	i. Leave salary & Pension contribution				-
	Total	1,00,00,000	32,50,000	28,00,000	27,77,088
	ii) Overtime allowance	-	-	-	-
	iii) Travelling allowance	12,00,000	7,00,000	7,00,000	6,34,409
2	Other charges (excluding major construction works)	60,00,000	87,00,000	82,00,000	81,80,683
3	Buildings and other works	1,03,00,000	1,13,50,000	47,00,000	46,89,561
	Grand Total	2,75,00,000	2,40,00,000	1,64,00,000	1,62,81,741

5.2 Human Resources

S.No	Name of the post	Sanctioned strength	Posts filled as on 31-3-96
1.	Scientific	43	18
2.	Technical	44	42
3.	Administrative	49	45
4.	Auxiliary	26	26
5.	Supporting	38	38
Total		200	169

5.3 Publications

5.3.1 Research Papers / Review Papers / Popular Articles

- 1 Balaguru, T. 1995. Programmed lessons and audio-tutorials for individualizing learning in Agriculture. *Journal of IAUA* 5 : 10-17.
- 2 Balaguru, T. and Manikandan, P. 1995. Co-training as a methodology for improving training effectiveness. *Indian Journal of Training and Development* XXV (3): 113.
- 3 Behera, U.K., Gadewar, A.U. and Satpathy, S. 1994. Scientists' perception about different facts of agricultural research projects. *Maharashtra J. Ext. Education* XVIII, 91-96.
- 4 Gadewar, A.U. 1996. Participatory approaches and techniques for rural development in the book titled "Rural development".
- 5 Muralidhar, U., Gadewar, A.U. and Kalla, J.C. 1996. Restructuring information system for effective decision making in agriculture and rural development sector. *Computer Society of India, A.P. Chapter*.
- 6 Muralidhar, U. and Rama Rao, D. 1996. Agricultural Universities Information System - Experiences of issues in Indian context. *Journal of Information Science*, Aug 1996.
- 7 Mehan, V.K., Reddy, P.M., Donald, D.Mc. and Vidyasagar Rao, K. 1995. Comparison of whole plant and detached leaf method for studying component of rust resistance in groundnut. *ICRISAT Arachin News*. 15, 25-28.
- 8 Rama Rao, D., Kalla J.C. and Muralidhar, U. 1995. Financing SAUs in India (in 2 parts). *Financial Agriculture*.
- 9 Samanta, R.K. and Prasad, M.V. 1995. An indigenous post harvesting technology. *Indigenous knowledge and Development Monitor*, 3 (2) 12-13.
- 10 Subudhi, H.N. and Samanta, R.K. 1995. Mangroves in India - Problems and prospects. *Environment news* 5 (6).

11. Sabarathnam, VE., 1996. Estimation of technological needs and identification of problems of farmers for formulation of research and extension programmes in agricultural entomology. *Experimental Agriculture*, 32 (1) 87-90.

5.3.2 Books / Reports

1. Anwer, M.M., Challa, J. and Shanmugam, S. 1995. Strategies of Human Resources Management. (Ed.) NAARM, Hyderabad.
2. Balaguru, T., Manikandan, P. and Kalla, J.C. 1996. Management Information System for Agricultural Research. NAARM, Hyderabad.
3. Gadewar, A.U. and Prasad, C. 1995. Participatory rural appraisal - concepts and applications in participatory approaches to sustainable rural development. (EQ). India Potash Ltd., New Delhi.
4. Routray, P., Samanta, R.K. and Prasad, M.V. 1995. Fisheries co-operatives: A new vistas for rural development in India in rural economy and development planning for the future. (Ed). Commonwealth publishers, New Delhi.
5. Rao, K.V.S., Kumar K.V. and Rama Rao, D. 1995. Manual on Forecasting packages. Software manual.

5.3.3 Book Reviews

Balaguru, T. 1995. Animal Quarantine in Asia and the Pacific - Report of an APO Seminar. Asian Productivity Organization, Tokyo. *Productivity* 35(4) : 750-751.

Sabarathnam, VE. Managing innovation concepts and tasks by N.S. Srinivasan and V. Narayanan. *Productivity* 35(4):752-53.

5.3.4 Technical Bulletins

Balaguru, T. 1996. Training: Is it a panacea for all performance problems of employees in organizations? NAARM, Hyderabad.

5.3.5 Papers presented in Seminars / Conferences / Workshops

1. Balaguru, T. and Vidyasagar Rao, K. Training of agricultural scientists in the use of computers. Paper presented in the 11th National Conference of Agricultural Research Statisticians held at Himachal Pradesh Krishi Viswa Vidyalaya, Palampur during October 16 to 18, 1995.

2. Balaguru, T. Farming Systems research for sustainable agricultural development in India. Paper presented in the 83rd Indian Science Congress held at Pattala during January 3-8, 1996.

3. Gadewar, A. U. Recent advances in data collection tools and techniques. Paper presented in the training-cum-workshop on 'Project formulation through participatory rural appraisal' held at the Gandhigram Rural Institute from November 28 to December 2, 1995.

4. Kalla, J.C. and Gopalam, A. Partnership in teacher development for a new Asia. Paper presented in the international conference organized by UNESCO and held at Bangkok, Thailand, from December 6 to 8, 1995.

5. Sabarathnam, V.E. and Pandey, A.K. Transfer of fisheries technologies in adopted and non-adopted villages of Puri district, Orissa. Paper presented in the National symposium on prospects and problems of technology transfer and rural reality organised by Bloved Research Society from February 17 to 18, 1996.

5.4 Visitors

A large number of visitors comprising national and international scientists from various organizations, students from various universities, trainees, and officers from different departments visited the Academy, who were apprised of the Academy's activities.

1. A group of 99 students belonging to College of Agricultural Extension, Annamalai University, led by Dr Santha Govind, Lecturer, visited on 04.04.1995, as a part of their study tour.
2. Mr Sudhir Krishna, Divisional Commissioner, Gulbarga Division, Karnataka, on 26-04-1995.
3. A group of postgraduate students, belonging to the Dept. of P.G. Studies and Research in Life Science, Kuvempu University, B.R. Project, visited on 15.5.1995 and 16.05.1995.
4. A group of 24 B.Sc. Final year Horticultural students, belonging to College of Horticulture, Dr Yashwant Singh Parmar University of Horticulture & Forestry, visited on 03.06.1995 and 05.06.1995 as a part of their study tour.
5. A group of 26 trainees of 71st Batch of Soil Conservation training of Central Soil and Water Conservation Research and Training Institute, Research Centre, Ooty, with their accompanying Technical Officer Mr. Jacob, visited on 25.07.1995.
6. Dr Balram Jakhar, Union Minister for Agriculture visited on 29-07-1995.
7. Dr M. Balasubramanian, Agricultural Research Specialist from World Bank, International Bank for Reconstruction and Development, International Development Association, visited on 01.08.1995.
8. A six-member delegation comprising senior officials like Commissioner of Agriculture in Kabbi State; Col. S.T. Bells, Chairman of APEC & Military administration, Kabbi State; Mrs. Jamila P. Abdullah, Regional Manager, FOCH; Mr. Alh Iff Abdullah, Prog. Manager, KARDA; Mr. Usman Saidu Gulma, Dy. Project Manager, NATSP, and Mr. Alh. Ahuhakar Bells, Dy. Project manager, KARDA, visited on 30.08.1995.
9. A group of 74 final year B.V.Sc. students accompanied by their staff members from the Veterinary college and research institute, TANVASU, Namakkal, visited on 06.09.1995, as a part of their educational tour.

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07.1995.

29-07-1995.

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missioner of Ag-
y administration,
. Alih Iff Abdullah,
anager, NATSP,
on 30.08.1995.

eir staff members
nakkal, visited on



Dr Jagdeesh C. Kalla, Director, NAARM, interacting with foreign delegates



Dr Norman Borlaug, Father of Green Revolution, (at the centre) and Prof. Timothy Reeves, DG, CIMMYT, (at the extreme left) are seen alongwith other eminent Indian agricultural scientists

10. A group of 15 visitors from the African and Asian countries, who were taking part in a training programme conducted at the National Institute of Small Scale Industries and Educational Training (NISIET), Hyderabad, led by Mrs. Rani visited on 20.09.1995.
11. A batch of 70 B.V.Sc. students of Madras Veterinary College, TNVASU, accompanied by Dr. P. Kanakaraj and Dr. V.S. Subramanian, Assoc. Professor, visited on 25.09.1995.
12. A batch of 91 students of Final B.Sc. (Ag) from Annamalai University, led by Mr. S. Thyagarajan, Lecturer, visited on 26.09.1995.
13. Mr A.R. Sukumar, Collector, Adilabad, Andhra Pradesh; Mr. Baijendra Kumar, Collector, Lagar, Madhya Pradesh; Mr Rajeev Ranjan, Collector, Tiruchirapalli, Tamilnadu; and Mr R. Subrahmanyam, Collector, Hyderabad, visited on 26-09-1995.
14. A team of eight Extension Officers and two Research Officers from the Dept. of Agriculture, Sri Lanka, undergoing training at the International Crops Research Institute for Semi-Arid Tropics, visited on 12.10.1995.
15. Eight-member team, headed by Dr K.E. Karunatilake, D.D. FCRDI, Dept. of Agriculture, Sri Lanka, visited on 12-10-1995.
16. A team of 20 B.Sc. (Ag) students from Dayanand Agric. College, Ajmer, Rajasthan, alongwith their lecturer Prof. I.H. Sharma, visited on 17.10.1995, as a part of their study tour.
17. A group of 69 final year B.Sc. (Ag) students from the College of Horticulture, Kerala Agricultural University, Vellanikkara, Trichur, alongwith their Associate Professor Dr P.K. Rajeevan, Head, Dept. of Pomology and Floriculture, visited on 21.10.1995.
18. A group of senior police officers and trainees undergoing 15th Training administrators course at S.V.P. National Police Academy, Hyderabad, alongwith Dr A.K. Saxena, visited on 09.11.1995.
19. A group of 17 B.Sc. Forestry and Horticulture students from ASPEE College of Home Science, alongwith Dr K.A. Thakkur, Associate Professor from Gujarat, visited on 11.12.1995.



20. Dr Mohammed A. Magese, Director-General, Bangladesh Agricultural Research Institute, Bangladesh, visited on 12-12-1995.
21. A group of nine M.Sc. (Final) Home Science Extension students alongwith Dr V. Sreedar, Reader in Home Science from S.V. University, Tirupathi, visited on 13.12.1995, as a part of their educational tour.
22. Dr J.E. Iken, IARLT, O.A.U and Dr E.D. Imolehih, NCRI from Nigeria, visited on 19-12-1995.
23. A team of 20 students of B.Sc. (Forestry) from Dr Y.S. Parmar University of Horticulture and Forestry, College of Forestry, Nauni, Solan, led by Dr P.S. Thakur and Dr. (Mrs) Anju Thakur, visited on 07.01.1996, as a part of their educational tour.
24. Mr Costas S. Petrides, Minister of Agriculture, Natural Resources and Environment of Republic of Cyprus, visited on 31-01-1996.
25. B.Sc. (Ag) students from Agricultural College and Research Institute from Tamil Nadu Agricultural University visited in two batches on 09.02.1996 and on 19.02.1996.
26. A group 68 final year U.G. students from College of Rural Home Science, UAS, Dharwad, led by Dr Rama Naik, visited NAARM on 24.02.1996.
27. Mr. Sangay Duba, Renewable Natural Resources Centre Bajo, P.O. Wangduephodring, W.C. Bhutan, visited on 13.03.1996.
28. Mr. Salihu S. Abubakar, NAERIS/ABU - ZARIA, Nigeria, visited on 16.03.1996.

5.5 COMMITTEES

5.5.1 Research Advisory Committee

The Research Advisory Committee (RAC) of NAARM was constituted by the ICAR with the sole purpose of suggesting relevant research programmes in the area of Agricultural Research and Educational Management. The RAC comprises the following senior level scientists.

1	Dr Maharaj Singh Ex-DDG (Education) New Delhi.	Chairman
2	Deputy Director General (Education) ICAR, New Delhi	Member
3	Dr M.S. Chari Director, CTRI Rajahmundry	Member
4	Dr Harpal Singh Dean, G.B. Pant University Pantnagar	Member
5	Dr Y.L. Nene DDG, ICRISAT Hyderabad	Member
6	Dr Dharni P. Sinha President and MD, COSMODE Hyderabad	Member
7	Dr K.S. Ramesh Dean, ASCI Hyderabad	Member
8	Dr Anil K. Gupta IIM, Ahmedabad	Member
9	Dr Jagdeesh C. Kalla Director, NAARM Hyderabad	Member

10. Dr T. Balaguru
Senior Scientist, NAARM
Hyderabad

Member-Secretary

The first meeting of the RAC was held on November 21, 1995, at NAARM, Hyderabad. The following recommendations emerged from the deliberations in the meeting.

1. Since the main purpose of NAARM is to work for the efficiency of management of agricultural research, education, and technology transfer functions in the ICAR-SAU system, its mandate needs to be slightly recast with the objective of providing a clear focus on the training and research functions including their interrelationship.
2. The research activity at NAARM should be revamped by following the guidelines laid down by the ICAR. The whole process of project development and implementation should be streamlined by preparing the relevant RPFs and discussing them in the Institute SRC.
3. The NAARM should prepare a comprehensive list of training programmes for various levels of functionaries in the ICAR-SAU system. An appropriate range of training courses to meet the specific training requirements of various functionary groups should be developed by NAARM. This information could then be organized in the form of a training plan with necessary details, including the allocation of faculty time.
4. In order to provide adequate inputs for the initial development and periodical updating / upgrading of various training programmes, as well as to render policy level support for improving the management efficiency of the overall system, NAARM should undertake, on a continuing basis, specific research projects
5. Depending upon the availability of faculty time, NAARM should consider undertaking additional research in the following areas on a priority basis.
 - i) Strategic research planning in support of the research plan preparation by the various institutes of ICAR who are required to progressively emphasize strategic research.
 - ii) Career advancement of scientists in the ICAR system under the present circumstances.
 - iii) Establishment of revolving fund for meeting the operational expenditure of agricultural research institutions.

- iv) A desk study on the rationalization of mandates of the ICAR institutes.
- v) Restructuring of ICAR institutes with a view to improving their operational efficiency.

6. The NAARM should ensure that all its training programmes are adequately backed up by appropriate research programmes. The present lop-sided allocation of faculty time to training and the grossly inadequate time devoted to research should be corrected through appropriate readjustment in the current training programmes and diversion of resources to the development of new research programmes. At any time, at least 20 per cent of the faculty time should be reserved for research activities.

At the same time, immediate action should be initiated by the ICAR to fill up the faculty positions lying vacant at NAARM so that the Academy could pay adequate attention to its research function besides meeting the requirements of various demand-driven training programmes.

7. In the face of inadequate faculty resource, due care should be taken by NAARM to avoid any duplication of research efforts with the sister institute like the National Institute of Agricultural Extension Management (MANAGE) in the area of technology transfer in agriculture.

8. As a 'think tank' to ICAR, the Academy should take up a desk study on the rationalization of the mandates of various research institutes to identify the avoidable overlap that might exist in their functions, and make suggestions to overcome the possible mismatch between their mandates and actual functioning.

9. On a high priority basis, NAARM should undertake an appropriate research study to identify the causes for the deterioration of work culture in the ICAR system after the switch over to the UGC pattern and should come up with viable suggestions for corrective action.

10. The NAARM should ultimately develop a comprehensive training plan and a corresponding comprehensive research plan with a rational allocation of the available faculty resource.

5.5.2 Quinquennial Review Team

The Quinquennial Review Team reviews the performance of the Academy with regard to achievements of its objectives in the last five years. It also examines and makes recommendations for the future research and academic programmes. The team comprising the following members visited the Academy from April 24 to 29, 1995.

- | | | |
|----|---|------------------|
| 1. | Dr R. Dwarakinath
Former VC, University of Agric. Sciences
FAO, Advisor, Bangalore | Chairman |
| 2. | Prof. S.C. Mandal
Former VC, RAU
Ranchi, Bihar | Member |
| 3. | Prof. N.S. Ramaswamy
Director, "CARTMAN"
Koramangala, Bangalore | Member |
| 4. | Dr K.N. Nag
Former VC, Rajasthan Agric. University
Udaipur, Rajasthan | Member |
| 5. | Dr R.K. Patel
Former Director, NDRI
Karnal | Member |
| 6. | Dr Dharni P. Sinha
Co-ordinator
Association of Management Development
Institutions of South Asia (AMDISA)
Hyderabad | Member |
| 7. | Dr S.N. Saha
Principal Scientist
NAARM, Hyderabad | Member-Secretary |

5.5.3 Management Committee

The Academy is guided in planning, organizing and conducting various types of training programmes and research studies by the Management Committee. The Management Committee approves the programmes to be undertaken. The Academy ensures their implementation, both in terms of physical targets and schedules. The Management Committee also helps in formulation and consolidation of proposals for Five Year Plan and Annual Plan, and takes up periodical review of progress of development schemes. While examining these programmes periodically, it pinpoints bottlenecks, if any, and suggests remedial measures.

The 25th and 26th meetings of the Management Committee of the Academy were held on May 20 and December 16, 1995, respectively. The composition of the Management Committee of NAARM is as follows :

- | | | |
|----|--|-------------------|
| 1. | Dr Jagdeesh C. Kalla
Director, NAARM, Hyderabad | Chairman |
| 2. | Dr M.V. Rao
Vice Chancellor, APAU, Hyderabad | Member |
| 3. | Dr J.C. Katyal
Director, CRIDA, Hyderabad | Member |
| 4. | Dr Harpal Singh
Dean, P.G. Studies
G.B. Pant University of Agriculture &
Technology, Pantnagar, U.P. | Member |
| 5. | Mr V. Ananda Rao
Principal Secretary
Dept. of Agriculture, Govt. of A.P. | Member |
| 6. | Mr P.V. Eapachan
Finance & Accounts Officer
Central Research Institute for
Dryland Agriculture, Hyderabad | Member |
| 7. | Mr Mohd. Haneef Khatri,
General Secretary
Dist. Congress Committee
Rajasthan | General Secretary |

- | | | |
|-----|--|------------------|
| 8. | Ms. K. Bharathi Devi
Social worker, Kakinada | Member |
| 9. | Dr O.P Pareek
Director,
NRC for Arid Zone Horticulture
Bikaner, Rajasthan | Member |
| 10. | Dr D. Jha
Director, NCAP
IASRI Campus
New Delhi | Member |
| 11. | Dr Mangala Rai
ADG (P)
ICAR Krishi Bhavan
New Delhi | Member |
| 12. | Dr J.C. Katyal
Director, CRIDA,
Hyderabad | Member |
| 13. | Mr M. Suresh Kumar
Chief Administrative Officer
NAARM, Hyderabad | Member-Secretary |

5.5.4 Academic Council

The Academic Council oversees all the academic functions of NAARM. The following was the composition of the Academic Council during 1995-96. The Council held two meetings during this year.

- | | | |
|----|---|----------|
| 1. | Dr Jagdeesh C. Kalla
Director, NAARM | Chairman |
| 2. | Dr P. Kumar
In-charge, ARSP Unit | Member |
| 3. | Dr A.U. Gadewar
In-charge, TOT Unit | Member |

- | | | |
|----|---|------------------|
| 4. | Dr A. Gopalam
In-charge, EST Unit | Member |
| 5. | Dr M.M. Anwer
In-charge, HRD Unit | Member |
| 6. | Ms. Suguna Sri
In-charge, ID Unit | Member |
| 7. | Dr S.N. Saha
Professor In-charge,
Training Programmes | Member-Secretary |

5.5.5 Research Council

The Research Council deliberates on the research activities in the areas of agricultural research and education management. The Committee consists of the following members.

- | | | |
|----|---|----------|
| 1. | Dr Jagdeesh C. Kalla
Director, NAARM | Chairman |
| 2. | Dr V. Raghavendra Rao
Principal Scientist, NAARM | Member |
| 3. | Dr P. Kumar
Principal Scientist, NAARM | Member |
| 4. | Dr S.N. Saha
Principal Scientist | Member |
| 5. | Dr K. Venkateswarlu
Principal Scientist, NAARM | Member |
| 6. | Dr A. Gopalam
Sr. Scientist, NAARM | Member |
| 7. | Dr K. Purnachandra Rao
Sr. Scientist, NAARM. | Member |

8.	Dr K.M. Reddy Sr. Scientist, NAARM	Member
9.	Dr M.M. Anwer Sr. Scientist, NAARM	Member
10.	Dr. K. Vidyasagar Rao Sr. Scientists, NAARM	Member
11.	Dr P. Manikandan Sr. Scientist, NAARM	Member
12.	Dr Jagannadham Challa Sr. Scientist, NAARM	Member
13.	Dr S. Shanmugam Sr. Scientist, NAARM	Member
14.	Dr VE. Sabarathnam Sr. Scientist, NAARM	Member
15.	Dr D. Rama Rao Sr. Scientist, NAARM	Member
16.	Dr A.U. Gadewar Sr. Scientist, NAARM	Member
17.	Mr U. Muralidhar Systems Analyst, NAARM	Member
18.	Dr. T. Balaguru Sr. Scientist, NAARM	Member-Secretary

During 1995-96, four meetings were conducted to finalize the research programmes of the Academy. .

5.5.6 Official Language Implementation Committee

The Official Language Implementation Committee of the Academy provides periodical suggestions for implementation of Official Language Policy, and rules and regulations of Government of India that are issued from time to time. The Committee met four times during the year on June 30, September 30, December 30, 1995, and March 30, 1996, and discussed several issues relating to the implementation of Hindi language. The following were the members :

- | | | |
|-----|---|------------------|
| 1. | Dr Jagdeesh C. Kalla
Director | Chairman |
| 2. | Dr A. Gopalam
Sr. Scientist | Vice-Chairman |
| 3. | Dr D. Rama Rao
Sr. Scientist | Member |
| 4. | Mr M. Suresh Kumar
Chief Administrative Officer | Member |
| 5. | Mr B. Pandu Reddy
Finance & Accounts Officer | Member |
| 6. | Mr K.R. Prabhakar
Programme Officer (CCTV) | Member |
| 7. | Mr U. Muralidhar
Systems Analyst | Member |
| 8. | Mr R.V.V.S. Prakasa Rao
Editor | Member |
| 9. | Mr. Abhijit Bhattacharjee
Public Relations Officer | Member |
| 10. | Ms Jhansi Laxmi, Sr. P.A. | Member |
| 11. | Ms G. Aneeja, Asst. Editor | Member |
| 12. | Mr D. Venkateswarlu
Tech. Officer (Hindi) | Member-Secretary |

5.5.7 Training Cell

The training cell is responsible for all the training activities of the Academy. It maintains the record of various details of these programmes. Professor in-charge of the cell is also responsible for performance evaluation of the Foundation Course programmes. The training cell comprises the following personnel.

1. Dr S.N. Saha, Professor In-charge, Training Programmes
2. Mr P. Vijender Reddy, T-II-3
3. Ms Sarada Samanta, Senior Stenographer

During the year under report, the training cell continued to plan the foundation and specialized courses. In all 21 advanced and specialized courses were conducted which trained 372 personnel from SAUs and ICAR institutes. In addition to the advanced courses 80 ARS scientists, freshly recruited into the Agricultural Research Service completed their foundation training at the Academy.

5.6 Personnel

1. Dr Jagdeesh C. Kalla, Director

Agricultural Research Systems & Policies Unit

2. Dr P. Kumar, Principal Scientist & Unit In-Charge
3. Dr VE. Sabarathnam, Senior Scientist
4. Dr T. Balaguru, Senior Scientist
5. Dr K. Vidyasagar Rao, Senior Scientist
6. Ms A. Vijaya Lakshmi, Stenographer
7. Ms A. Mercy, Stenographer
8. Mr M. Shyam Rao, SSG-I

Educational Systems and Technology Unit

9. Dr A. Gopalam, Senior Scientist & Unit In-Charge
10. Dr K.M. Reddy, Senior Scientist
11. Dr D. Rama Rao, Senior Scientist
12. Mr M. Venkatesh, Stenographer
13. Mr M. Yadaiah, SSG-II

Transfer of Technology Systems & Policies Unit

14. Dr A.U. Gadewar, Senior Scientist & Unit-in-charge
15. Mr R.C. Chelvan, Senior Scientist (on study leave)
16. Mr N.R. Nageswara Rao, Technical Officer (T-5)
17. Mr L. Venkateswarlu, Photographer (T-4)
18. Mr Bansidhar Nayak, Projectionist (T-II-3)
19. Mr D. Rajagopal Rao, PAE Operator
20. Mr S. Sunder Raj, Media Operator (T-I)
21. Ms T. Vanisri, Stenographer
22. Mr G. V. Bikshapathi, SSG-II

Project Management Unit

23. Dr V. Raghavendra Rao, Principal Scientist & Unit-in-Charge (up to 03-02-96)
24. Dr K. Venkataeswarlu, Principal Scientist
25. Dr P. Manikandan, Senior Scientist
26. Ms L. Jhansi Lakshmi, Senior Personal Assistant
27. Ms K.K. Rukmani Ammal, Junior Stenographer
28. Mr K. Satyanarayana, SSG-II

Human Resources Development Unit

- 29. Dr M.M. Anwer, Senior Scientist & Unit In-Charge
- 30. Dr Jagannadham Challa, Senior Scientist
- 31. Dr S. Shanmugam, Senior Scientist
- 32. Ms S. Sessa Sai, Junior Stenographer
- 33. Mr N. Narasinga Rao, SSG-I

Information and Documentation Unit

- 34. Ms M. Suguna Sri, Librarian & Unit In-Charge
- 35. Ms P.V. Nirmala, Technical Officer (T-5)
- 36. Mr Y.V. Siva Prasad, Documentation Asst. (T-4)
- 37. Ms S. Shanti, Junior Stenographer
- 38. Mr Venkatesham, Xerox Operator
- 39. Mr M. Ashok, SSG-II

Director's Cell

- 40. Mr N. Raghunath, Stenographer
- 41. Mr T.V. Ram Das, Stenographer
- 42. Mr P. Swamy, SSG-II

Training Cell

- 43. Dr S. N. Saha, Professor & In-charge, Training Programmes
- 44. Ms Sarada Samanta, Senior Stenographer
- 45. Mr P.Vijender Reddy, Research Assistant (T-4)
- 46. Mr K. Hemlal, SSG-I

Hindi Section

- 47. Mr D. Venkateswarlu, Technical Officer (T-5)
- 48. Mr S. Pradeep Singh, Asst. Director (Hindi Instruction)
- 49. Ms J. Renuka, Hindi Assistant
- 50. Ms P. Raja Rajeswari Jyothi, Junior Clerk

Computer Laboratory

- 51. Mr U. Muralidhar, Programmer (Computer)
- 52. Mr M.J. Pradeep Kumar, Asst. Programmer

- 53. Mr K. Venkateswara Kumar, Statistical Asst. (T-4)
- 54. Mr P. Mohan Singh, Computer Assistant (T-II-3)

Video Instruction and Production Laboratory

- 55. Mr K.R. Prabhakar, Programme Officer (CCTV)
- 56. Mr Ch. Janardhan Rao, Technical Officer (T-5)
- 57. Mr N. Naresh Kumar, VCR-Tele Cine Operator (T-I-3)
- 58. Mr C. Bikshapathi, SSG-II

Production and Publication

- 59. Mr R.V.V.S.Prakasa Rao, Editor (T-7)
- 60. Mr N. Narasimha Reddy, Technical Officer (T-5)
- 61. Mr K. Anil Kumar, Reprographic Assistant (T-4)
- 62. Ms G. Aneja, Asst Editor (T-II-3)
- 63. Mr M. Shekar Reddy, Dark Room Assistant (T-II-3)
- 64. Mr P. Srinivas, Proof Reader
- 65. Mr S. Swamy, Assistant Gestetner Operator
- 66. Mr Prashant Gaikwad, Binder
- 67. Mr C. Chandramouli, SSG-I
- 68. Ms C. Narasubai, SSG-I

Farm Section

- 69. Mr M.A. Basith, Technical Officer (T-5) & Jr. Farm Supdt.
- 70. Mr K. Rangaswamy, Technical Officer (T-5)
- 71. Mr B. Veeraiah, Farm Assistant (T-I-3)
- 72. Mr Ahire Laxman Maharu, Hort. Tech. Assistant (T-II-3)
- 73. Mr K. Shivaiah, Technician Electrical (T-2)
- 74. Mr C.J. Samuel, Junior Clerk
- 75. Mr M. Padmaiah, Tractor Driver
- 76. Mr P. Iswari, Power Tiller Driver
- 77. Mr P. Balraj, SSG-II
- 78. Mr S. Jangaiah, SSG-II
- 79. Mr B. Subba Rao, SSG-II
- 80. Ms B. Bharatamma, SSG-II
- 81. Ms N. Sukunamma, SSG-I
- 82. Mr G. Pentalah, SSG-I
- 83. Mr R. Sattalah, SSG-I
- 84. Ms G. Anasuya, SSG-I

Hostel Services

85. Mr Zameer Ahmed, Manager (Hostel Services)
86. Mr Sham Bahadur, Catering In-Charge
87. Ms Savithri Srinivasan, Mess Supervisor
88. Mr P. Bhaktavastalam, Junior Clerk
89. Mr K. Pentalah, SSG-II
90. Mr P. Yadalah, SSG-II
91. Mr Khalid, SSG-II
92. Ms B. Santhamma, SSG-II
93. Ms S. Sakunthala, SSG-II
94. Mr A. Yadalah, SSG-I
95. Mr M. Krishnaiah, SSG-I
96. Mr C. Anjalah, SSG-I
97. Ms K. Kalavathi, SSG-II
98. Ms M. Yadamma, SSG-I
99. Ms I. Bharathamma, SSG-I
100. Ms T. Jangamma, SSG-I
101. Ms Nazia Parveen, SSG-I

Administration

102. Mr M. Suresh Kumar, Chief Administrative Officer
103. Mr. Mohd. Habeebuddin, Assistant Administrative Officer
104. Mr K. Janakiramaiah, Superintendent
105. Mr T. Ashok Kumar, Superintendent
106. Mr Y. Sankar Rao, Superintendent
107. Mr P.P. Brahmaji, Assistant
108. Mr P. Neelakantam, Assistant
109. Mr P.G. Kohad, Assistant
110. Mr M. Narasimha Rao, Assistant
111. Mr K. Prabhu Das, Senior Clerk
112. Mr T. Srinivas, Senior Clerk
113. Mr M. Dinesh, Senior Clerk
114. Mr G. Raj Reddy, Senior Clerk
115. Mr C. Phani Raj, Senior Clerk
116. Mr P. Anand Kumar, Stenographer
117. Ms B. Padma Saroja, Junior Clerk
118. Mr K. Srinivasa Rao, Junior Clerk
119. Mr P. Srinivasu, Junior Clerk
120. Mr M. Sridhar, Junior Clerk

121. Mr P. Venkatesh, Junior Clerk
122. Mr K.R. Ghanshyam, Junior Clerk
123. Mr M.K. Samson, Junior Clerk
124. Ms N. Vijaya Lakshmi, Junior Stenographer
125. Ms V. Shailaja, Junior Stenographer
126. Ms P.D. Jessie Ecclicia, PMBX Operator-cum-Receptionist
127. Mr P. Anantha Reddy, Technician (Instrument) (T-II-3)
128. Ms K.Swarajya Laxmi, Junior Library Assistant (T-2)
129. Mr B.H. Dharma Raj, SSG-III
130. Mr J. Chandraiah, SSG-I

Public Relations Cell

131. Mr Abhijit Bhattacharjee, Public Relations Officer
132. Mr A. Murali Mohan Rao, Technical Officer (T-5)
133. Mr P. Namdev, Technical Officer (T-5)
134. Ms Y. Anuradha, Jr. Stenographer

Civil Engineering Cell

135. Mr Sohail Ahmad Khan, Junior Engineer(Civil) (T-4)
136. Mr K. Veeranasiah, Carpenter
137. Mr N. Prabhakar Rao, Plumber

Transport Section

138. Mr K. Sattaiah, Driver
139. Mr S.K. Peera, Driver
140. Mr M.M. Mohan Rao, Vehicle Mechanic (T-I-3)
141. Mr G. Muthyalu, Driver
142. Mr U.V. Ratnam, Driver
143. Mr N. Ashok, Driver
144. Mr T. Laxman, Driver
145. Mr B. Narayana, SSG-III

Security Wing

146. Mr B.Ch. Satyanarayana, Estate-cum-Security Officer
147. Mr M. Srinivasa Rao, Pump Driver
148. Mr B.K. Venkat Ram, Pump Driver
149. Mr G. Dasaratha, SSG-I
150. Ms M. Eswaramma, SSG-I

- 151. Mr M. Phool Kumar, SSG-I
- 152. Mr L. Satyanarayana, SSG-I

Electrical Wing

- 153. Mr L. Ramesh, Technical Assistant (Elec.) (T-I-3)
- 154. Mr K. Obulapathi, Technician Electrical (T-2)
- 155. Mr M.K. Sonkusare, Electrician (T-2)
- 156. Mr M. Mohan Rao, Technician Electrical (T-2)
- 157. Mr B.S.N. Murthy, Technician Electrical (T-2)
- 158. Mr M.K. Shamsuddin, Technician Electrical (T-2)
- 159. Mr A. Nageswara Rao, Lineman (T-2)
- 160. Mr B. Satyanarayana, Technician Electrical (T-1)
- 161. Mr D.R.S. Rao, Pump Driver

Audit and Accounts Section

- 162. Mr B. Pandu Reddy, Finance & Accounts Officer
- 163. Mr C. Bagaiah, Assistant
- 164. Mr S. Balakamesh, Junior Clerk
- 165. Mr R. Chandra Babu, Junior Clerk
- 166. Ms K. Radha Sujatha, Junior Stenographer
- 167. Ms Y. Gayathri, Junior Clerk
- 168. Ms G. Manibai, SSG III

Health Centre

- 169. Dr (Major) A. Debnath, Medical Officer
- 170. Mr A. Venkateswarlu, Compounder

Appointments

S.No.	Name and	Designation
1.	Dr K.M. Naidu	Joint Director (Retd. on 31-05-96)
2.	Mr Pradeep Singh	Assistant Director (Hindi Instruction)
3.	Mr Laxman Mahru Ahire	Hort. Tech. Assistant (T-II-3)
4.	Mr K. Pentaiah	Cook
5.	Mr P. Yadaiah	Cook
6.	Ms C. Narsu Bai	SSG-I
7.	Ms Nikita Gopal	Research Associate, DST Project
8.	Mr B.V.Devendra Rao	Technical Assistant, DST Project

Promotions

S.No.	Name	From	To
1.	Mr R.V.V.S. Prakasa Rao	Editor (T-6)	Editor (T-7)
2.	Mr Y. Shankara Rao	Assistant	Superintendent
3.	Mr M.A. Basith	Lab. Tech (T-4)	Technical Officer (T-5)
4.	Mr A. Murali Mohan Rao	Artist (T-4)	Technical Officer (T-5)
5.	Mr Ch. Janardhan Rao	Video Cameraman	Technical Officer (T-5)
6.	Mr P. Namdev	Graphic-cum- Visualisor	Technical Officer (T-5)
7.	Mr K. Rangasamy	Farm Assistan	Technical Officer (T-5)
8.	Mr P.Vijender Reddy	Museum Asst	Research Asst. (T-4)
9.	Mr N. Naresh Kumar	VCR - Tele Cine Operator (T-2)	VCR - Tele Cine Operator (T-1-3)
10.	Mr M. Sekhar Reddy	Dark Room Asst	Dark Room Asst. (T-1-3)
11.	Mr M.K. Sonkusare	Technician. Elec	Technician Elec. (T-2)
12.	Mr K. Shivaiah	Technician Elec	Technician Elec. (T-2)
13.	Mr A. Nageswara Rao	Technician Elec	Technician Elec. (T-2)
14.	Mr M. Mohan Rao	Technician Elec	Technician Elec. (T-2)
15.	Ms B. Bharatamma	SSG-I	SSG-II
16.	Ms T. Vanisri	Jr. Stenographer	Stenographer
17.	Mr T.V. Ram Das	Jr. Stenographer	Stenographer
18.	Mr B.H. Dharma Raj	SSG-III	Junior Clerk

Deputations

1. Dr R.K. Samanta, Sr. Scientist, NAARM as Director, MANAGE
2. Dr K.P.C. Rao, Sr. Scientist, NAARM as Director, MANAGE

Transfer on Promotion

1. Mr T. Ashok Kumar, Superintendent, NAARM, Hyderabad, as Asst. Adm. Officer, NRCO, Yeluru, Andhra Pradesh
2. Mr Mohd. Habeebuddin Asst. Admn. Officer, NAARM, as Administrative Officer, NRCS, Hyderabad
3. Ms M. Girija Kumari, Jr. Stenographer, NAARM, as Superintendent, A&A PDP, Hyderabad

Deaths

1. Dr V. Raghavendra Rao, Principal Scientist
2. Ms D. Sangamma, Cook



We record, with profound grief, the demise of Dr V. Raghavendra Rao, Principal Scientist, who passed away on February 4, 1996. He was 57 and is survived by wife, a son, and a daughter.

As the Head of the Project management unit of the Academy, Dr Rao trained scientists of ARS in modern techniques of management in relation to planning, scheduling, and control of agricultural research projects. He organized training programmes on agricultural research project management for senior scientists, heads of ICAR institutes, and professors of SAUs. Dr Rao also prepared case materials and conducted workshops on various aspects of project management. One of his major contributions is the development of rating system for appraising agricultural research projects.

Dr Rao did his M.Sc. in Mycology and plant pathology (1960) and Ph.D. in plant pathology (1966) from the Indian Agricultural Research Institute (IARI), New Delhi. He attended an executive course on general management at the Administrative Staff College of India during 1980. He visited France to study the Agricultural education and research systems during 1982; United States of America in 1988 to study the U.S. Agricultural Research Service systems. In the U.S., he also participated in the course on project implementation for agricultural and rural development, and attended some courses on project management, conflict management, and organizational behaviour at the George Washington University, Washington D.C.

Earlier Dr Rao served as a Nematologist at the Indian Institute of Horticultural Research, Bangalore, from 1969 to 1977. He had published various articles on plant pathology, and on plant parasitic nematodes and their control.

Expressing shock and sorrow on his unfortunate and sudden demise, the faculty and staff met on February 5, 1996, and passed a condolence resolution, which was communicated to his family. Dr Raghavendra Rao's memory will always be treasured by all those who had the privilege of knowing his sunny disposition, his fine mind, and his affectionate heart.

